

CERTIFICATE

This is to certify that the project titled “**Handwritten Digit Recognition Using Convolutional Neural Networks (CNN)**” has been successfully completed and submitted as part of the academic requirement.

Guide: Mr.Aditya Bangia Sir

Student Name: Ayush Tiwari

Date: 27 Nov 2025

ACKNOWLEDGEMENT

I express my sincere gratitude to my guide, teachers, and institution for providing support and guidance throughout this project. I am also thankful to my friends and family for their encouragement, without which this project would not have been possible.

OBJECTIVES

1. To design and implement a CNN model for handwritten digit recognition.
2. To understand convolution, pooling, and dense layers in deep learning.
3. To train and evaluate the model on the MNIST dataset.
4. To analyze accuracy, loss, and confusion matrix.
5. To study applications, advantages, and limitations of CNN-based digit recognition.

SYNOPSIS

This project aims to build a Convolutional Neural Network (CNN) for recognizing handwritten digits using the MNIST dataset. The system follows preprocessing, model building, training, evaluation, and prediction steps. CNN automatically extracts features such as edges and curves, enabling high accuracy. This technique is widely used in banking, postal automation, and digital form processing.

DETAILED PROJECT REPORT

Handwritten Digit Recognition Using Convolutional Neural Networks (CNN)

1. Introduction

Handwritten digit recognition is one of the most widely researched topics in the fields of machine learning, pattern recognition, and computer vision. It involves the automatic identification of handwritten numerical digits (0–9) from images. This technique plays a crucial role in real-world applications such as postal mail sorting, bank cheque verification, form processing, digitizing handwritten notes, and automating manual data entry.

Earlier, handwritten digit recognition was solved using traditional machine learning algorithms like k-NN and SVM. These methods required manual feature extraction such as detecting edges and shapes. However, handwriting styles vary significantly, and these models often failed on unseen writing patterns. Deep learning, especially Convolutional Neural Networks (CNNs), revolutionized this field by automatically learning features directly from pixel data, achieving state-of-the-art results.

2. Dataset Description

The MNIST dataset is used for training and testing. It contains 60,000 training and 10,000 testing images, each 28×28 pixels and grayscale. Although simple, MNIST contains enough variation (thick strokes, slanted digits, noise) to test model robustness.

3. Data Preprocessing

Before training, images must be normalized (0–255 → 0–1), reshaped to (28, 28, 1), and labels converted into one-hot encoded vectors for multi-class classification.

4. CNN Architecture

The model includes:

- Convolutional layers with 32 and 64 filters (3×3, ReLU)
- Max-pooling layers (2×2)
- Flatten layer
- Dense layer with 128 neurons (ReLU)
- Output layer with 10 neurons (Softmax)

This architecture effectively learns digit features such as strokes, curves, and shapes.

5. Model Training

The model is trained using:

- Loss: Categorical Crossentropy
- Optimizer: Adam
- Batch size: 32
- Epochs: 10–20

CNNs converge quickly on MNIST due to its structured dataset. A well-trained model achieves 98–99% testing accuracy.

6. Evaluation and Results

The results are measured using accuracy, loss, and confusion matrix. Digits like 4 and 9 may be confused due to similar shapes. Overall performance remains high, demonstrating excellent generalization.

7. Applications

CNN-based digit recognition is used in banking (cheque scanning), postal services (PIN extraction), education (OMR reading), data entry automation, and document digitization.

8. Advantages

- Very high accuracy
- No manual feature extraction

- Robust to handwriting variations
- Scalable to full OCR systems

9. Limitations

- Requires large training datasets
- Needs GPU for fast training
- Struggles with noisy backgrounds or handwriting styles not seen in training

10. Future Scope

Future improvements include extending to full text OCR, deploying on mobile devices, using advanced architectures (ResNet), and combining CNN with RNN/LSTM for word-level recognition.

11. Conclusion

A CNN model for handwritten digit recognition was successfully implemented with high accuracy. The project demonstrates the power of deep learning for image classification and lays the foundation for advanced OCR systems.